

FitBuddy: AI Fitness Plan Generator using Gemini Models

Project Description

FitBuddy is a web-based application that uses Generative AI to create **personalized 7-day workout plans and nutrition tips** for weight loss, muscle gain or general wellness. A FastAPI backend takes validated user details (name, age, weight, goal and workout intensity), sends them to Google Gemini, stores the generated plan in SQLite, and lets users submit feedback to refine the plan with AI. A short nutrition or recovery tip is attached to every plan, and an admin view lists all users with their original and updated plans.

Scenarios

Scenario 1: Personalized Workout Plan. The user enters name, age, weight, fitness goal (for example weight loss or muscle gain) and preferred intensity (high, medium or low). Gemini generates a day-by-day 7-day plan with a warm-up, main workout and cooldown for each day, tailored to the goal and experience level.

Scenario 2: Feedback-Based Plan Update. The user submits feedback on the existing plan, such as "more focus on cardio" or "include more rest days". The original plan and the feedback are sent to Gemini, which returns a revised plan that is stored alongside the original.

Scenario 3: Nutrition and Recovery Tip. The user receives a concise tip based on the selected goal, for example "Include protein in your post-workout meal" for muscle gain. The tip complements the workout plan and is shown below it.

Scenario 4: Admin and Coach Dashboard. An admin or coach opens the View All Users page to see every registered user with their details, original plan and updated plan, so progress can be tracked and plans compared across multiple clients.

Technical Architecture

Three layers: an HTML front end rendered with Jinja2 templates, a FastAPI back end (`main.py`, `routes.py`, `schemas.py`, `database.py`) and external services (Google Gemini Pro and Flash) with SQLite and SQLAlchemy for storage. See 3. *Project Design Phase / Solution Architecture* for the diagram.

Pre-requisites

1. Python 3.10+ - <https://www.python.org>
2. FastAPI - <https://fastapi.tiangolo.com>
3. Google AI Studio account and Gemini API key - <https://aistudio.google.com/apikey>
4. SQLAlchemy and SQLite - <https://docs.sqlalchemy.org>
5. Jinja2 templates - <https://jinja.palletsprojects.com>
6. Uvicorn ASGI server - <https://www.uvicorn.org>
7. Git and a GitHub account
8. Basic HTML and CSS

Project Workflow

Milestone	Activities	Where in the code
1. Model Selection & Architecture	1.1 Research and select the Gemini models; 1.2 define the application architecture; 1.3 set up the development environment	requirements.txt, .env, app/main.py
2. Core Functionalities Development	2.1 Workout plan, nutrition tip, feedback update, storage and admin view; 2.2 FastAPI routing and user input processing	gemini_generator.py, gemini_flash_generator.py, updated_plan.py, database.py, schemas.py
3. routes.py Development	3.1 Core routes /, /generate-workout, /submit-feedback, /view-all-users and the JSON API routes	routes.py
4. Frontend Development	4.1 Designing the user interface; 4.2 dynamic Jinja2 templates	templates/, static/images/
5. Deployment	5.1 Prepare the application for local deployment; 5.2 test and verify the local deployment	requirements.txt, .env, app/main.py

Milestone 1: Model Selection and Architecture

1. Review the project needs: structured 7-day plans, short nutrition tips, plan updates from feedback and fast web responses.
2. Compare options. Hugging Face models (T5, BART, DistilBERT, Bloom) needed local hosting or GPU and heavy prompting, so Google **Gemini** was chosen.
3. Use Gemini 1.5 Pro for workout generation and plan updates, and Gemini Flash for fast nutrition tips.
4. Create a virtual environment: `python -m venv fitbuddy-env`, then activate it.
5. Install the libraries: `pip install fastapi uvicorn jinja2 sqlalchemy python-multipart google-generativeai`.
6. Run the server with `uvicorn app.main:app --reload` and open `http://127.0.0.1:8000` (API docs at `/docs`).

Note: Gemini 1.5 Pro is used for workouts and updates and Gemini Flash for tips. Check Google AI Studio for the model names currently available before running the project.

Milestone 2 and 3: Backend

- `database.py` stores users and plans with SQLAlchemy (`save_user`, `save_plan`, `update_plan`, `get_original_plan`, `get_user`); `schemas.py` holds the Pydantic models `UserInput` and `FeedbackRequest`.
- `gemini_generator.py` builds the workout prompt and calls Gemini Pro; `gemini_flash_generator.py` returns the nutrition tip; `updated_plan.py` revises a plan from user feedback.
- `routes.py` exposes the routes: `/`, `/generate-workout`, `/submit-feedback`, `/view-all-users`, `/generate-workout/gemini`, `/nutrition-tip`, `/generate-plan` and `/update-plan/{user_id}`. Interactive API docs are at `/docs`.

Milestone 4: Frontend Development

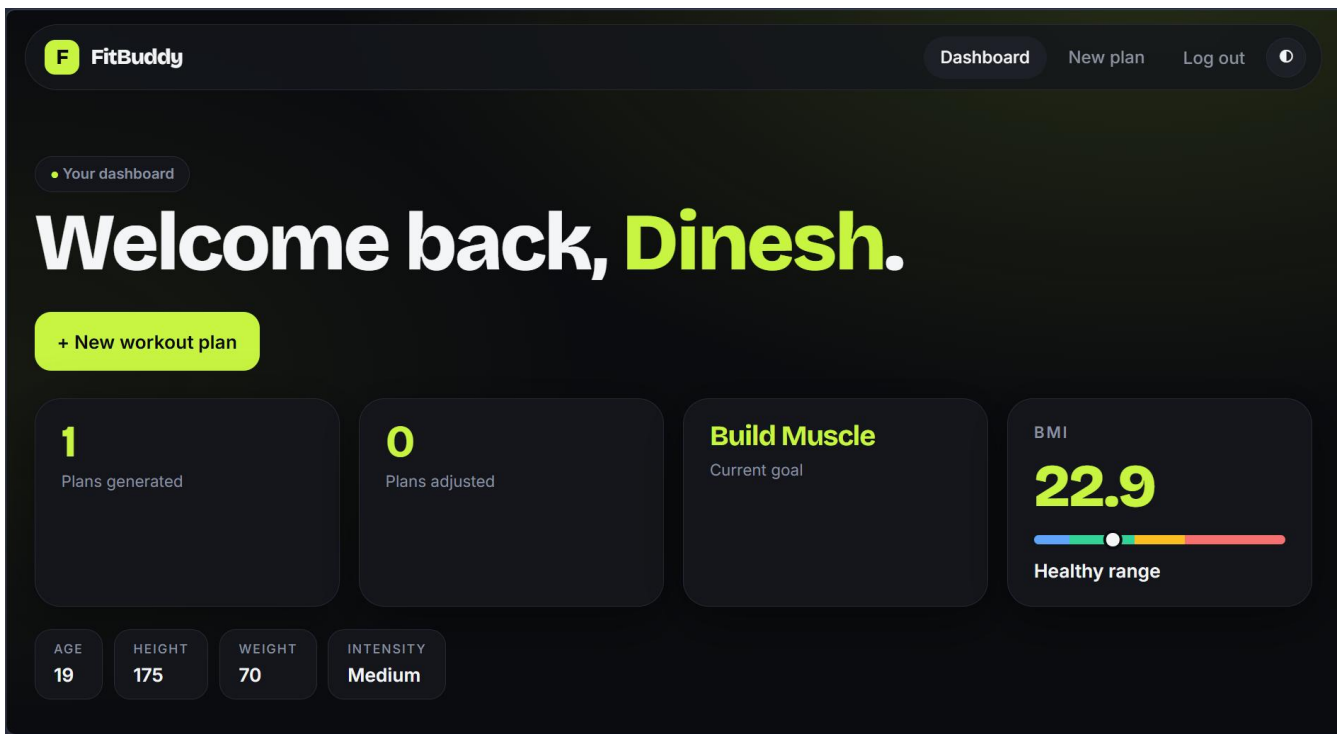
- `index.html` is the input form with name, user ID, age, weight, goal and intensity; `result.html` shows the plan, nutrition tip and feedback form; `all_users.html` is the admin panel.

- Pages use embedded CSS, Flexbox, media queries and a gym-themed background image; Jinja2 TemplateResponse passes the plan and tip to the templates.

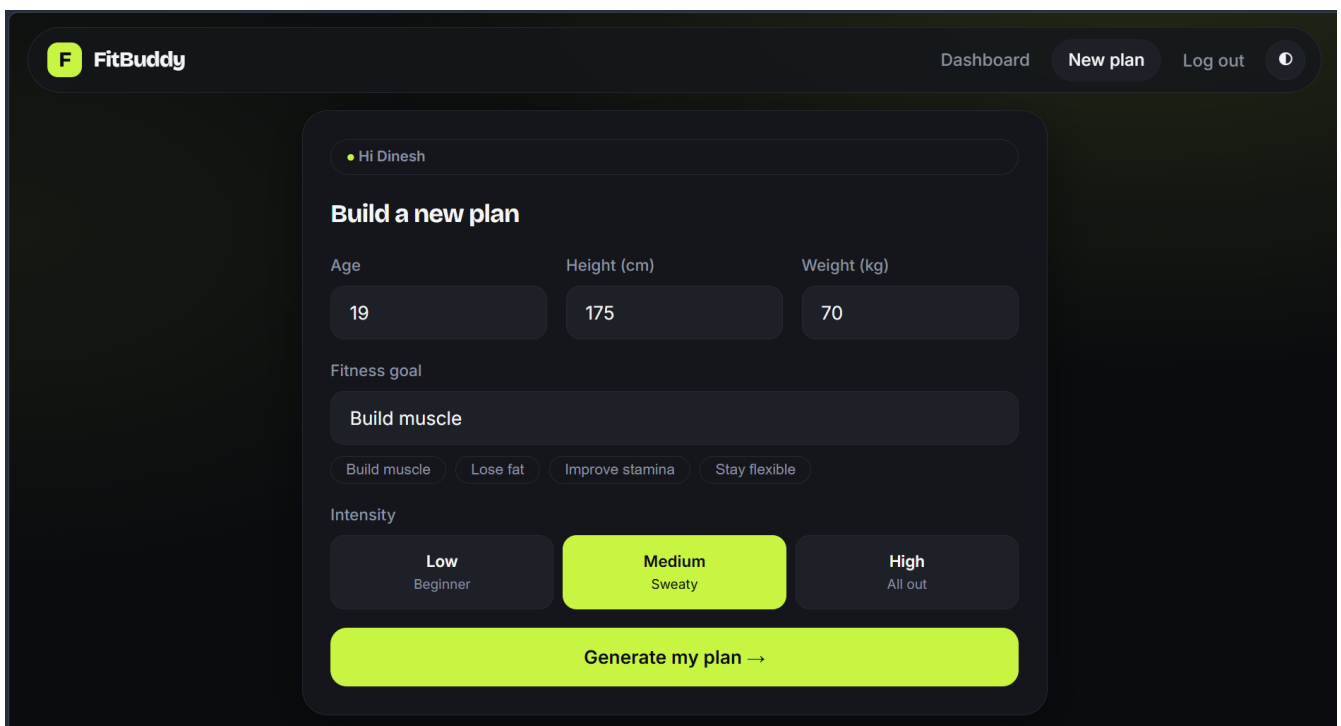
Milestone 5: Deployment

The application runs locally with FastAPI and Uvicorn. Install `requirements.txt` in a virtual environment, store the key in `.env` as `GOOGLE_API_KEY`, start the server with `uvicorn main:app --reload` and test each page.

Home page



Personalized workout generation page



Workout plan

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FitBuddy

Dashboard

New plan

Log out

0

Hey **Dinesh**, let's get to work.

GOAL

INTENSITY

AGE

HEIGHT

WEIGHT

Build Muscle

Medium

19

175

70

7-day workout plan

Copy

Print / PDF

7-Day Backup Workout Plan

This plan is a temporary fallback for goal: Build muscle at medium intensity.

Day 1 (Full Body)

• Warm up: 8 min brisk walk + arm circles

• Main: Squats 3×12, Push-ups 3×10, Glute bridges 3×15, Plank 3×30s

• Cooldown: Hamstring + chest stretch 5 min

Day 2 (Cardio + Core)

• Warm up: 5 min light jog

BODY MASS INDEX

22.9

Healthy range

Estimate assuming weight in kg. BMI is a rough guide only.

NUTRITION & RECOVERY

Fallback tip: prioritize protein at each meal and maintain hydration; retry in 1-2 minutes for an AI-personalized tip.

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FitBuddy

Dashboard

New plan

Log out

0

Day 3 (Lower Body)

• Warm up: Dynamic leg swings 5 min

• Main: Lunges 3×12/leg, Step-ups 3×12/leg, Calf raises 3×20

• Cooldown: Quad + calf stretch 5 min

Day 4 (Recovery)

• Warm up: Easy walk 10 min

• Main: Mobility flow 20 min + light yoga

• Cooldown: Deep breathing 5 min

Day 5 (Upper Body)

• Warm up: Shoulder rolls + band pull-aparts 5 min

• Main: Incline push-ups 3×12, Rows 3×12, Overhead press 3×10

• Cooldown: Shoulder + tricep stretch 5 min

Day 6 (Conditioning)

• Warm up: Jumping jacks 3 min

• Main: 6 rounds (30s work / 30s rest): squats, mountain climbers, plank

ADJUST WITH AI

More cardio, add yoga...

Update plan

Day 6 (Conditioning)

- Warm up: Jumping jacks 3 min
- Main: 6 rounds (30s work / 30s rest): squats, mountain climbers, plank
- Cooldown: Full body stretch 8 min

Day 7 (Active Rest)

- Warm up: Gentle walk 5 min
- Main: 20-30 min light activity (walk, cycle, or yoga)
- Cooldown: Stretch and hydration

FitBuddy · AI-generated plans are guidance, not medical advice.

Nutrition tip

NUTRITION & RECOVERY

Fallback tip: prioritize protein at each meal and maintain hydration; retry in 1-2 minutes for an AI-personalized tip.

Feedback page

ADJUST WITH AI

More cardio, add yoga...

Update plan

Conclusion

FitBuddy combines FastAPI with Google Gemini to deliver personalized 7-day workout plans and nutrition or recovery tips. Users enter their details once, receive a structured plan, and can refine it through feedback while SQLite keeps both the original and updated versions. The admin view gives coaches a clear picture of every user, making FitBuddy suitable for individuals as well as trainers managing several clients.